Grade 11 Term 2 Home Practice – C1

Parameters and sample statistics

Global Economics – 2026–2027

Contents

1 Introduction 1

2 Notation Reference and practice procedure 2

3 General Worked problems 3

4 Advanced Problems 13

Evaluated criteria

C1 Distinguishes population parameters from sample statistics by interpreting and comparing the notation for population and sample means, standard deviations, and proportions in economic and financial contexts.

How to practise. Try each task before following its blue solution link. Use the arrows to return.

1 Introduction

Problems:

1.1. From electricity bills to a mean

1.2. Quick check: redraw or redefine?

1.1 From electricity bills to a mean

A utility studies the amount charged per bill. Its **target population** is all May electricity bills for small shops in one district. Its **sample** consists of 40 randomly selected bills from that list.

The **population parameter** μ is the unknown mean charge across all bills in the list. The **sample statistic** $\bar{x} = 210$ thousand COP is the mean charge across the 40 selected bills and estimates μ.



1.2 Quick check: redraw or redefine?

The utility draws 40 bills again, either from the same May list or from a combined May list covering small shops in two districts. Which population mean does each draw estimate?

Suggested answer. A redraw can give a different $\bar{x}_{may}$ while estimating the same μ_{may} for the original district. The combined draw's mean $\bar{x}_{2D}$ estimates μ_{2D}, the population mean across both districts. Those two population means could have the same numerical value.



2 Notation Reference and practice procedure

Problems:

2.1. Notation reference
2.2. Practice procedure

2.1 Notation reference

Population parameter	Sample statistic	Units
μ, mean of the variable across all units	$\bar{x}$, mean across the selected units	Same as the variable
σ, standard deviation of the variable across all units	s, standard deviation across the selected units	Same as the variable
p, fraction of all units meeting a condition	$\hat{p}$, fraction of selected units meeting that condition	Unitless, as a fraction, decimal or percentage

The mean describes the average level, and the standard deviation describes how spread out the values are. In the utility example, μ and σ describe the amount charged, in thousand COP, across the specified May bills. A different month, district or group of customers changes which bills define the parameters. Measuring electricity use on the same bills would instead give a mean and standard deviation in kilowatt-hours. For a proportion, specify the condition as well as the population. For instance, p could be the fraction of those May bills paid by the due date.



2.2 Practice procedure

1. Name the unit being studied and define the target population, including any time, place or eligibility limits. Then identify the selected units and the sample size.
2. State the variable and its units for a mean or standard deviation. For a proportion, state the condition and identify the denominator as the size of the group being described. To find $\hat{p}$, divide the number of selected units meeting the condition by the sample size.
3. Match the quantity to its symbol. Ask whether it summarizes the whole target population or the selected units. Write the contextual meaning even when the numerical value is unknown.
4. Before subtracting, check that the variable or condition and the units match. Name the two groups and say whether the result compares two sample statistics, a sample statistic with a population parameter, or two known population parameters. Express differences between percentages in percentage points.
5. For a redraw from the same population with the same measures, the parameters stay fixed and the statistics can vary. If the target population changes, redefine the parameters for the new group. A difference between sample statistics alone cannot establish the size or direction of a difference between population parameters.



3 General Worked problems

Problems:

3.1. A market association's sales survey (C1)

3.2. A supermarket's checkout survey (C1)

3.3. A transit agency's fare review (C1)

3.4. A coffee association's export shipments (C1)

3.5. A credit union's loan review (C1)

3.1 A market association's sales survey (C1)

A market association wants to describe all 1,200 stalls licensed for June. From 60 randomly selected stalls, it finds a mean June sales revenue of 4.8 million COP and a sample standard deviation of 1.2 million COP. Eighteen sampled stalls accept digital payment.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the population mean, standard deviation and proportion, and pair them with the corresponding sample statistics.

C1 To check how the survey results vary across samples, the association draws another random sample from the same June list. It finds a mean of 5.1 million COP, a standard deviation of 1.0 million COP and a digital-payment share of 0.25. Revisit your Task 1 answers. What changes, what stays the same, and which numerical differences can you calculate?

Sample (green)	**Population (red)**
Sample mean ($\bar{x}$)	Population mean (μ)
Sample standard deviation (s)	Population standard deviation (σ)
Sample proportion ($\hat{p}$)	Population proportion (p)
Sample (selected units)	Population (all units of interest)

C1 – Match each quantity to its symbol

The market association randomly selects 60 stalls from all 1,200 June-licensed stalls. June sales average 4.8 million COP with a standard deviation of 1.2 million COP. Eighteen sampled stalls accept digital payment, a share of 30%. These statistics estimate μ, σ and p, respectively.

Population parameter	Sample statistic
μ, mean June sales across all licensed stalls	$\bar{x}_1 = 4.8$ million COP, mean across the 60
σ, SD of June sales across all licensed stalls	$s = 1.2$ million COP, SD across the 60
p, fraction of licensed stalls accepting digital payment	$\hat{p} = 18/60 = 0.30$, fraction among the 60

Sales measures use million COP. Payment proportions are unitless.



C1 – Compare what changes across samples

The association first draws 60 stalls from all 1,200 June-licensed stalls, then makes a fresh random draw from the same June list. In draw order, June sales means are 4.8 and 5.1 million COP, standard deviations are 1.2 and 1.0 million COP, and shares accepting digital payment are 30% and 25%.

Quantity	Original scenario	Revised scenario	What the change tells us
Population	All 1,200 June-licensed stalls	Same June list	μ, σ and p stay fixed.
Sample	First 60 selected stalls	Fresh draw from the June list	A new draw can produce different statistics.
Mean	$\bar{x}_1 = 4.8$ million COP	$\bar{x}_2 = 5.1$ million COP	$\bar{x}_2 - \bar{x}_1 = 5.1 - 4.8 = +0.3$ million COP.
Standard deviation	$s = 1.2$ million COP	$s = 1.0$ million COP	$1.0 - 1.2 = -0.2$ million COP in s.
Digital-payment share	$\hat{p} = 18/60 = 0.30$	$\hat{p} = 0.25$	$0.25 - 0.30 = -0.05$ in $\hat{p}$ (-5 percentage points).



3.2 A supermarket's checkout survey (C1)

A supermarket manager wants to describe all 1,500 purchases made at one store on Saturday. In a random sample of 50 receipts, the mean purchase value is 64 thousand COP and the sample standard deviation is 18 thousand COP. Twenty of those purchases used a loyalty card.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the population mean, standard deviation and proportion, and pair them with the corresponding sample statistics.

C1 Before reporting the results, the manager asks a colleague to draw another random sample of 50 receipts from that store's Saturday records. The new sample has a mean purchase value of 61 thousand COP and a standard deviation of 20 thousand COP. Twenty-five of these purchases used a loyalty card. Revisit your Task 1 answers. What changes, what stays the same, and which numerical differences can you calculate?

C1 – Match each quantity to its symbol

To study all 1,500 Saturday purchases at one store, the manager randomly selects 50 receipts. Purchase values average 64 thousand COP with a standard deviation of 18 thousand COP. Twenty sampled purchases used a loyalty card, a share of 40%. These statistics estimate the population mean μ, standard deviation σ and proportion p.

Population parameter	Sample statistic
μ, mean value of all Saturday purchases	$\bar{x}_1 = 64$ thousand COP, mean among the 50 purchases
σ, standard deviation of value across all Saturday purchases	$s = 18$ thousand COP, standard deviation among the 50 purchases
p, fraction of all Saturday purchases that used a loyalty card	$\hat{p} = 20/50 = 0.40$, fraction among the 50 purchases

C1 – Compare what changes across samples

The manager and a colleague each draw 50 receipts from the store's same 1,500 Saturday purchases. From the first draw to the second, the mean purchase value goes from 64 to 61 thousand COP and the standard deviation goes from 18 to 20 thousand COP. Loyalty-card counts of 20 and 25 give sample shares of 40% and 50%.

Quantity	Original scenario	Revised scenario	What the change tells us
Population	All 1,500 Saturday purchases at the store	Same 1,500 purchases	μ, σ and p stay fixed.
Sample	First draw of 50 receipts	Second draw of 50 receipts	The sample size stays the same. The draw is new.
Mean	$\bar{x}_1 = 64$ thousand COP	$\bar{x}_2 = 61$ thousand COP	$\bar{x}_2 - \bar{x}_1 = 61 - 64 = -3$ thousand COP.
Standard deviation	$s = 18$ thousand COP	$s = 20$ thousand COP	$20 - 18 = +2$ thousand COP in s.
Loyalty-card share	$\hat{p} = 20/50 = 0.40$	$\hat{p} = 25/50 = 0.50$	$0.50 - 0.40 = +0.10$ in $\hat{p}$ ($+10$ percentage points).

3.3 A transit agency's fare review (C1)

A transit agency wants to describe all 8,400 bus boardings on its central routes last Monday. In a random sample of 100 boarding records, the mean fare paid is 3.2 thousand COP and the sample standard deviation is 0.8 thousand COP. Thirty-five of the sampled boardings used a discounted fare.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the population mean, standard deviation and proportion, and pair them with the corresponding sample statistics.

C1 For a fare review, the agency's director asks to include suburban routes alongside the central routes, using the same Monday records. The combined routes have 12,000 boardings. A random sample of 100 from the combined records gives a mean fare of 3.4 thousand COP and a standard deviation of 0.9 thousand COP. Thirty sampled boardings used a discounted fare. Revisit your Task 1 answers. What changes, what stays the same, and which numerical differences can you calculate?

C1 – Match each quantity to its symbol

The transit agency randomly selects 100 records from all 8,400 Monday boardings on the central routes. Fares average 3.2 thousand COP with a standard deviation of 0.8 thousand COP. Thirty-five sampled boardings used a discounted fare, a share of 35%. These statistics estimate the population mean μ_{ctr}, standard deviation σ and proportion p.

Population parameter	Sample statistic
μ_{ctr}, mean fare paid across all central-route boardings that Monday	$\bar{x}_{\text{ctr}} = 3.2$ thousand COP, mean among the 100 boardings
σ, standard deviation of fares across all those boardings	$s = 0.8$ thousand COP, standard deviation among the 100 boardings
p, fraction of all those boardings that used a discounted fare	$\hat{p} = 35/100 = 0.35$, fraction among the 100 boardings



C1 – Compare what changes when routes are added

The original draw includes 100 boardings from the 8,400 central-route boardings on Monday. After adding suburban routes for that Monday, the agency draws 100 new records from the 12,000 combined boardings. Mean fares are 3.2 and 3.4 thousand COP, with standard deviations of 0.8 and 0.9 thousand COP. Discounted fares account for 35 and 30 boardings, or 35% and 30%.

Quantity	Original scenario	Revised scenario	What the change tells us
Population	8,400 Monday boardings on central routes	12,000 Monday boardings on central and suburban routes	The population gains $12{,}000 - 8{,}400 = 3{,}600$ boardings. The day stays the same.
Sample	100 central-route boardings	100 boardings from the combined routes	The sample size stays at 100. The pool it is drawn from expands.
Mean	μ_{ctr} for central routes, $\bar{x}_{\text{ctr}} = 3.2$ thousand COP	μ_{comb} for combined routes, $\bar{x}_{\text{comb}} = 3.4$ thousand COP	$\bar{x}_{\text{comb}} - \bar{x}_{\text{ctr}} = 3.4 - 3.2 = +0.2$ thousand COP. The difference between the population means is unknown.
Standard deviation	σ for central routes, $s = 0.8$ thousand COP	σ for combined routes, $s = 0.9$ thousand COP	$0.9 - 0.8 = +0.1$ thousand COP in s. Whether σ changed is unknown.
Discounted-fare share	p for central routes, $\hat{p} = 35/100 = 0.35$	p for combined routes, $\hat{p} = 30/100 = 0.30$	$0.30 - 0.35 = -0.05$ in $\hat{p}$ (-5 percentage points). Whether p changed is unknown.



3.4 A coffee association's export shipments (C1)

A coffee exporters' association wants to describe all 900 coffee shipments in its April export records. In a random sample of 40 shipments, the mean shipment weight is 18.4 metric tonnes and the sample standard deviation is 4.0 metric tonnes. Twelve of the sampled shipments departed on schedule.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the population mean, standard deviation and proportion, and pair them with the corresponding sample statistics.

C1 A second analyst repeats the study with a fresh random draw of 40 shipments from the association's April records. The mean shipment weight is 19.1 metric tonnes and the standard deviation is 3.6 metric tonnes. Twelve of these shipments departed on schedule. Revisit your Task 1 answers. What changes, what stays the same, and which numerical differences can you calculate?

C1 – Match each quantity to its symbol

The coffee exporters' association randomly selects 40 shipments from all 900 April coffee shipments. Shipment weights average 18.4 metric tonnes with a standard deviation of 4.0 metric tonnes. Twelve sampled shipments departed on schedule, a share of 30%. These statistics estimate the population mean μ, standard deviation σ and proportion p.

Population parameter	Sample statistic
μ, mean weight across all 900 April shipments	$\bar{x}_1 = 18.4$ metric tonnes, mean among the 40 shipments
σ, standard deviation of weight across all 900 April shipments	$s = 4.0$ metric tonnes, standard deviation among the 40 shipments
p, fraction of all 900 April shipments that departed on schedule	$\hat{p} = 12/40 = 0.30$, fraction among the 40 shipments



C1 – Compare what changes across samples

The two analysts each draw 40 shipments from the same 900 April shipments. In draw order, mean shipment weights are 18.4 and 19.1 metric tonnes, and standard deviations are 4.0 and 3.6 metric tonnes. Each draw contains 12 departures on schedule, so both on-schedule shares are 30%.

Quantity	Original scenario	Revised scenario	What the change tells us
Population	All 900 shipments in the April records	Same 900 shipments	μ, σ and p stay fixed.
Sample	First draw of 40 shipments	Second draw of 40 shipments	The sample size stays the same. The draw is new.
Mean	$\bar{x}_1 = 18.4$ metric tonnes	$\bar{x}_2 = 19.1$ metric tonnes	$\bar{x}_2 - \bar{x}_1 = 19.1 - 18.4 = +0.7$ metric tonnes.
Standard deviation	$s = 4.0$ metric tonnes	$s = 3.6$ metric tonnes	$3.6 - 4.0 = -0.4$ metric tonnes in s.
On-schedule share	$\hat{p} = 12/40 = 0.30$	$\hat{p} = 12/40 = 0.30$	$0.30 - 0.30 = 0$ in $\hat{p}$ (0 percentage points).



3.5 A credit union's loan review (C1)

A credit union wants to describe all 800 of its active microenterprise working-capital loans on June 30. In a random sample of 50 loans, the mean outstanding balance is 8.0 million COP and the sample standard deviation is 2.4 million COP. Ten of the sampled loans have an overdue installment on that date.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the population mean, standard deviation and proportion, and pair them with the corresponding sample statistics.

C1 The lending committee asks the analyst to include active microenterprise equipment loans in the June 30 review. The two loan categories together contain 1,000 loans. In a random sample of 50 from the combined list, the mean outstanding balance is 8.8 million COP and the standard deviation is 2.9 million COP. Eight sampled loans have an overdue installment on that date. Revisit your Task 1 answers. What changes, what stays the same, and which numerical differences can you calculate?

C1 – Match each quantity to its symbol

The credit union randomly samples 50 loans from all 800 active microenterprise working-capital loans on June 30. Outstanding balances average 8.0 million COP with a standard deviation of 2.4 million COP. Ten sampled loans have an overdue installment that day, a share of 20%. These statistics estimate the population mean μ_{wc}, standard deviation σ and proportion p.

Population parameter	Sample statistic
μ_{wc}, mean outstanding balance across all 800 loans	$\bar{x}_{\text{wc}} = 8.0$ million COP, mean among the 50 loans
σ, standard deviation of outstanding balances across all 800 loans	$s = 2.4$ million COP, standard deviation among the 50 loans
p, fraction of all 800 loans with an overdue installment on June 30	$\hat{p} = 10/50 = 0.20$, fraction among the 50 loans



C1 – Compare what changes when a loan category is added

The credit union first draws 50 loans from the 800 active working-capital loans on June 30. It then draws 50 loans from the 1,000 active loans in the combined working-capital and equipment list for that date. Mean balances are 8.0 and 8.8 million COP, with standard deviations of 2.4 and 2.9 million COP. The overdue counts of 10 and 8 give shares of 20% and 16%.

Quantity	Original scenario	Revised scenario	What the change tells us
Population	800 active working-capital loans	1,000 active working-capital and equipment loans	The population gains $1{,}000 - 800 = 200$ loans. The review date stays the same.
Sample	50 working-capital loans	50 loans from the combined list	The sample size stays at 50. The pool it is drawn from expands.
Mean	μ_{wc} for working-capital loans, $\bar{x}_{\text{wc}} = 8.0$ million COP	μ_{comb} for both loan categories, $\bar{x}_{\text{comb}} = 8.8$ million COP	$\bar{x}_{\text{comb}} - \bar{x}_{\text{wc}} = 8.8 - 8.0 = +0.8$ million COP. The difference between the population means is unknown.
Standard deviation	σ for working-capital loans, $s = 2.4$ million COP	σ for both loan categories, $s = 2.9$ million COP	$2.9 - 2.4 = +0.5$ million COP in s. Whether σ changed is unknown.
Overdue share	p for working-capital loans, $\hat{p} = 10/50 = 0.20$	p for both loan categories, $\hat{p} = 8/50 = 0.16$	$0.16 - 0.20 = -0.04$ in $\hat{p}$ (-4 percentage points). Whether p changed is unknown.



4 Advanced Problems

Problems:

4.1. A retailer expands its fee review (C1)
4.2. A payment processor checks its July report (C1)

4.1 A retailer expands its fee review (C1)

A retailer's finance team is preparing to renegotiate its card-processing contract. It wants to know the average fee per payment and how much fees vary, so it can budget for processing costs. It also tracks the average purchase amount to describe the size of the payments behind those fees. High fees flag potentially costly payments, and early refunds help the team plan for cash reversals.

The team links September card-payment records to processor statements and refund logs. It records the processing fee and original purchase amount in thousand COP, flags a fee of at least 2 thousand COP as high, and checks whether each payment was refunded within seven calendar days. The refund logs cover the full seven-day window for each payment. The review starts with the 2,400 in-store payments. In a random sample of 60, the mean fee is 1.4 thousand COP, the fee standard deviation is 0.6 thousand COP and the mean purchase amount is 80 thousand COP. Twelve payments have a high fee and six are refunded within seven days.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the two population means, the fee standard deviation and the two proportions. Pair each parameter with its corresponding sample statistic and calculate both sample proportions.

C1 The team adds 1,600 online payments to cover both sales channels in September. It draws 80 payments at random from the combined records, using the same measures and definitions. The mean fee is 1.8 thousand COP, the fee standard deviation is 0.7 thousand COP and the mean purchase amount is 95 thousand COP. Twenty-four sampled payments have a high fee and 12 are refunded within seven days. Revisit your Task 1 answers. What changes, what stays the same, and which differences can you calculate?

C1 – Match the retailer's parameters and statistics

The retailer randomly samples 60 payments from all 2,400 in-store card payments in September. In the sample, the processing fee averages 1.4 thousand COP with a standard deviation of 0.6 thousand COP, and the original purchase amount averages 80 thousand COP. Twelve payments have a fee of at least 2 thousand COP (20%), and six are refunded within seven calendar days (10%). These statistics estimate the population fee mean $\mu_{F,I}$ and standard deviation $\sigma_{F,I}$, mean purchase amount $\mu_{A,I}$, high-fee share $p_{H,I}$ and refund share $p_{R,I}$.

Population parameter	Sample statistic
$\mu_{F,I}$, mean processing fee across all in-store payments	$\bar{x}_{F,I} = 1.4$ thousand COP, mean fee among the 60
$\mu_{A,I}$, mean original purchase amount across all in-store payments	$\bar{x}_{A,I} = 80$ thousand COP, mean purchase amount among the 60
$\sigma_{F,I}$, fee standard deviation across all in-store payments	$s_{F,I} = 0.6$ thousand COP, fee standard deviation among the 60
$p_{H,I}$, fraction of all in-store payments with a fee of at least 2 thousand COP	$\hat{p}_{H,I} = 12/60 = 0.20$, high-fee share among the 60
$p_{R,I}$, fraction of all in-store payments refunded within seven calendar days	$\hat{p}_{R,I} = 6/60 = 0.10$, seven-day refund share among the 60

The first subscript identifies the measure and I identifies the in-store review. The sample statistics estimate the parameters, whose values are unknown.



C1 – Compare the in-store and combined reviews

The original review draws 60 payments from 2,400 in-store payments in September. The revised review draws 80 payments from all 4,000 payments across both channels in the same month. In order, the sample mean fees are 1.4 and 1.8 thousand COP, with standard deviations of 0.6 and 0.7 thousand COP. Mean purchase amounts are 80 and 95 thousand COP. The high-fee shares are 20% and 30%, and the seven-day refund shares are 10% and 15%.

Quantity	Original scenario	Revised scenario	What the change tells us
Population	2,400 in-store payments	4,000 combined payments	The review adds $4{,}000 - 2{,}400 = 1{,}600$ online payments. The month and measure definitions stay the same.
Sample	60 payments from the in-store list	80 payments from the combined list	The pool expands and the sample size increases by $80 - 60 = 20$.
Mean fee	$\mu_{F,I}$ $\bar{x}_{F,I} = 1.4$ thousand COP	$\mu_{F,C}$ $\bar{x}_{F,C} = 1.8$ thousand COP	$\bar{x}_{F,C} - \bar{x}_{F,I} = 1.8 - 1.4 = +0.4$ thousand COP
Mean purchase amount	$\mu_{A,I}$ $\bar{x}_{A,I} = 80$ thousand COP	$\mu_{A,C}$ $\bar{x}_{A,C} = 95$ thousand COP	$\bar{x}_{A,C} - \bar{x}_{A,I} = 95 - 80 = +15$ thousand COP
Fee standard deviation	$\sigma_{F,I}$ $s_{F,I} = 0.6$ thousand COP	$\sigma_{F,C}$ $s_{F,C} = 0.7$ thousand COP	$0.7 - 0.6 = +0.1$ thousand COP
High-fee share	$\hat{p}_{H,I} = 12/60 = 0.20$ $p_{H,I}$	$\hat{p}_{H,C} = 24/80 = 0.30$ $p_{H,C}$	$30 - 20 = +10$ percentage points
Seven-day refund share	$\hat{p}_{R,I} = 6/60 = 0.10$ $p_{R,I}$	$\hat{p}_{R,C} = 12/80 = 0.15$ $p_{R,C}$	$15 - 10 = +5$ percentage points

The subscript C refers to the combined review. Its parameters summarize a broader population. Each measure's numerical difference in the last column compares sample statistics. Exact differences between parameters would require summaries of all payments in each target population.



4.2 A payment processor checks its July report (C1)

A payment processor is checking its July service report for merchants in one city. It uses the card-transaction register, payout ledger and verification log to trace each payment through settlement. Settlement time runs from payment until the merchant receives the funds and is measured in business days. The net amount credited to the merchant after processing fees is measured in thousand COP. The team tracks time to assess service speed and net credits to describe how much cash merchants receive per payment. It flags a settlement as late if it takes more than two business days, and separately flags any transaction that required manual verification before release to track staff workload.

From all July card transactions for its merchants in that city, the processor randomly selects 50. The mean settlement time is 1.6 business days, and the mean net credit is 145 thousand COP. Five settlements are late and eight transactions required manual verification.

Questions/Tasks



C1 Identify the population and sample. Give the symbols and contextual meanings for the two population means and two population proportions. Pair each parameter with its corresponding sample statistic and calculate both sample proportions.

C1 A colleague takes a fresh random sample of 50 transactions from the same July register, using the same definitions. The mean settlement time is 1.9 business days and the mean net credit is 152 thousand COP. Nine settlements are late and six transactions required manual verification. Revisit your Task 1 answers. What changes, what stays the same, and how do the two draws differ? The complete June records for the same city give benchmarks of 1.8 business days, 150 thousand COP, 16% late and 12% requiring manual verification. Compare each July draw with June and explain what would establish the exact month-to-month differences.

C1 – Match the processor's parameters and statistics

The processor randomly selects 50 transactions from all July transactions for its merchants in one city. Settlement takes an average of 1.6 business days and the mean net credit after processing fees is 145 thousand COP. Five settlements take more than two business days (10%), and eight transactions require manual verification before release (16%). These statistics estimate the July mean settlement time $\mu_{T,J}$, mean net credit $\mu_{N,J}$, late share $p_{L,J}$ and manual-verification share $p_{M,J}$.

Population parameter	Sample statistic
$\mu_{T,J}$, mean settlement time across all July transactions	$\bar{x}_{T,J,1} = 1.6$ business days, mean time among the 50
$\mu_{N,J}$, mean net amount credited across all July transactions	$\bar{x}_{N,J,1} = 145$ thousand COP, mean net credit among the 50
$p_{L,J}$, fraction of all July transactions taking more than two business days to settle	$\hat{p}_{L,1} = 5/50 = 0.10$, late share among the 50
$p_{M,J}$, fraction of all July transactions requiring manual verification before release	$\hat{p}_{M,1} = 8/50 = 0.16$, manual-verification share among the 50

The first subscript identifies the measure. For means, J denotes July and 1 or 2 identifies the draw. For proportions, J denotes the July population and 1 denotes the first draw. The July parameters are unknown.



C1 – Compare the July draws and check the June benchmarks

The first and second draws each use 50 transactions from the same July register for the city. In draw order, mean settlement times are 1.6 and 1.9 business days, and mean net credits are 145 and 152 thousand COP. The late shares are 10% and 18%, and the manual-verification shares are 16% and 12%. The June benchmarks come from all June transactions for the same city's merchants (Jun in the mean subscripts).

Quantity	Original scenario	Revised scenario	What the change tells us
Population	All July card transactions in scope	Same July card transactions	$\mu_{T,J}$, $\mu_{N,J}$, $p_{L,J}$ and $p_{M,J}$ stay fixed.
Sample	First draw of 50 transactions	Second draw of 50 transactions	The draw changes. The sample size and measures stay the same.
Mean settlement time	$\bar{x}_{T,J,1} = 1.6$ business days	$\bar{x}_{T,J,2} = 1.9$ business days	$\bar{x}_{T,J,2} - \bar{x}_{T,J,1} = 1.9 - 1.6 = +0.3$ business days
Mean net credit	$\bar{x}_{N,J,1} = 145$ thousand COP	$\bar{x}_{N,J,2} = 152$ thousand COP	$\bar{x}_{N,J,2} - \bar{x}_{N,J,1} = 152 - 145 = +7$ thousand COP
Late share	$\hat{p}_{L,1} = 5/50 = 0.10$	$\hat{p}_{L,2} = 9/50 = 0.18$	$18 - 10 = +8$ percentage points
Manual-verification share	$\hat{p}_{M,1} = 8/50 = 0.16$	$\hat{p}_{M,2} = 6/50 = 0.12$	$12 - 16 = -4$ percentage points

First July draw minus June population. Mean settlement time differs by $\bar{x}_{T,J,1} - \mu_{T,Jun} = 1.6 - 1.8 = -0.2$ business days and mean net credit by $\bar{x}_{N,J,1} - \mu_{N,Jun} = 145 - 150 = -5$ thousand COP. The late share differs by $10 - 16 = -6$ percentage points and the manual-verification share by $16 - 12 = +4$ percentage points.

Second July draw minus June population. The differences in the same order are $\bar{x}_{T,J,2} - \mu_{T,Jun} = 1.9 - 1.8 = +0.1$ business days, $\bar{x}_{N,J,2} - \mu_{N,Jun} = 152 - 150 = +2$ thousand COP, $18 - 16 = +2$ percentage points and $12 - 12 = 0$ percentage points.

The table compares statistics from two July draws. The June comparisons subtract known population parameters from July sample statistics. To find exact month-to-month differences, compute the four July parameters from the complete July records or use verified July population summaries, then subtract the corresponding June parameters.

